Quantitative Financial Risk Management

Duisenberg Honours Programme in Quantitative Risk Management

Period 4.5

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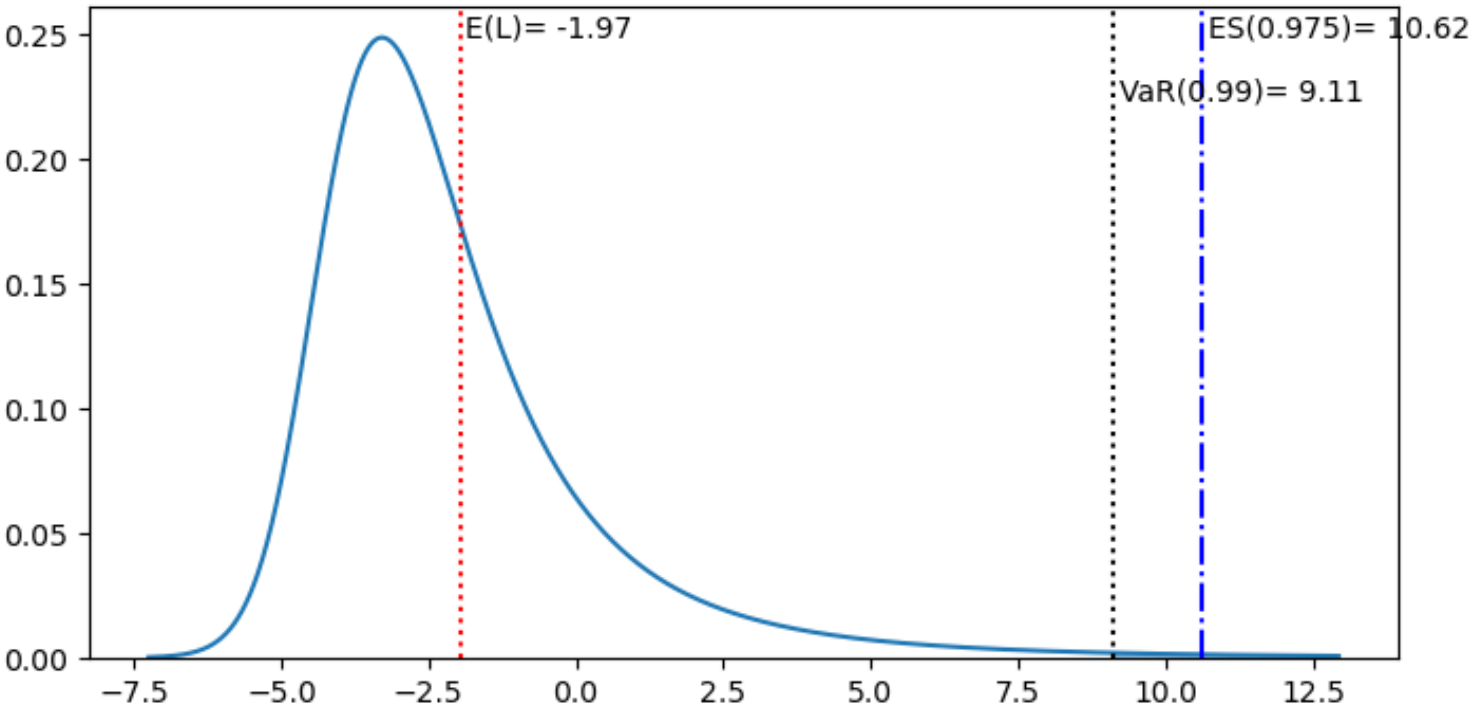
Part II

Market Risk Measurement

(McNeil, Frey, and Embrechts, 2015, Ch. 2, Part I)

1. Motivation
2. Portfolio Values and Losses
3. Unconditional Risk Estimation
4. Tails and distributions
5. Conditional Risk Management
6. Review

What Risk? How do we measure them?



Main risk measures for portfolios or risky assets are:

- VaR (Value-at-Risk) at horizon h, confidence level $\alpha\%$: highest loss with probability α when holding portfolio for h days
- ES or CVaR (Expected Shortfall/Conditional Value-at-Risk) at horizon h, confidence level $\alpha\%$: average or expected loss when VaR is exceeded

How to estimate these measures for a portfolio of risky assets, from historical data of prices/returns of these risky assets?

- These risk measures are uniquely determined by the distribution of losses, which we will call loss distribution

(MFE §2.3)

Consider

- portfolio with value V_t,
- observable at time t
- looking at risk over $[t, t+1]$
- where $V_{t+1|t}$ is unknown/random.
- $L_{t+1} = -(V_{t+1} - V_t)$ is the loss, with loss distribution,
- and we call $-L_{t+1}$ the profit, or profit-and-loss, with P&L distribution.

(note we look at *loss= direct difference* here, not log-differences or percentage differences)

Or, formally:

$$\mathrm{VaR}_\alpha = q_\alpha(F_L) = F_L^{\leftarrow}(\alpha) \equiv F_L^{-1}(\alpha) \quad \text{Value-at-Risk} \tag{2.1}$$

$$\mathrm{ES}_\alpha = E(L|L > \mathrm{VaR}_\alpha) \quad \text{Expected Shortfall} \tag{2.2}$$

$$= \frac{1}{P(L > \mathrm{VaR})} \int_{\mathrm{VaR}}^{\infty} z f_L(z) dz$$

which means:

- VaR: What amount could I lose maximally with prob α?
- ES: If things go wrong, what is my expected loss? What is the average loss if VaR is exceeded?

If we suppose that $F_L(l) \equiv F_{L,t}(l)$, all losses drawings from i.i.d. loss distribution F_L, then

- VaR and ES do not change from day to day
- hence we can do unconditional risk estimation
- and use historical data to estimate loss distribution and/or risk measures
- which is done typically for credit or operational risk

Nice and simple, such an assumption!

However, conditions change (dramatically!), as do the risks...

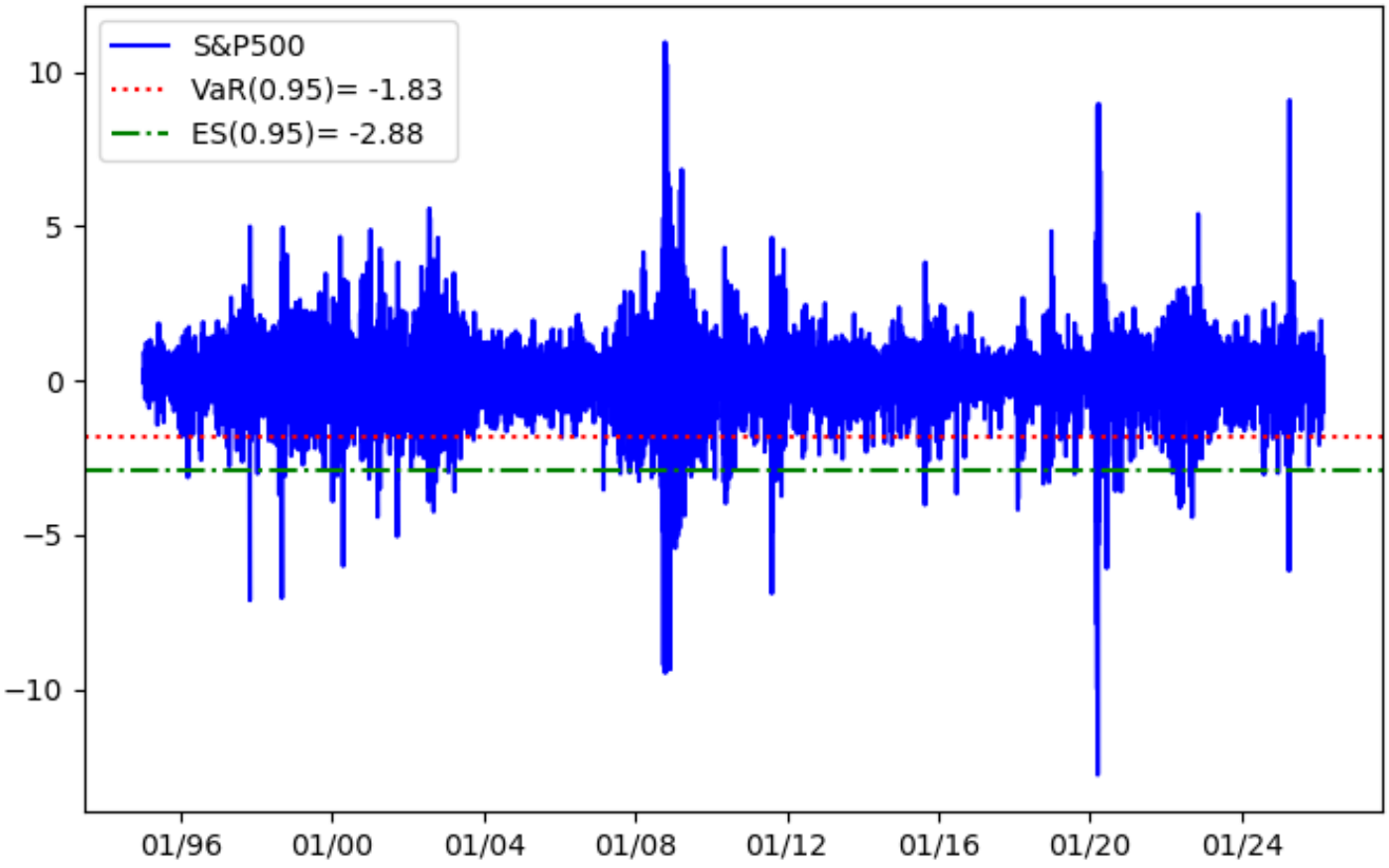
- Hence better to do conditional risk estimation
- assuming losses have a distribution $F_{L,t+1}(l|\mathcal{F}_t)$, changing day-to-day e.g. due to changing volatility
- which depends on the history of risk factors covered in $\mathcal{F}_t$

Effectively, we look for

$$L_{t+1} = l_{[t]}(\boldsymbol{X}_{t+1}|\mathcal{F}_t)$$

= conditional predictive loss for tomorrow, dependent on unknown outcome X_{t+1} of the state of affairs, given current risk factors $\mathcal{F}_t$.
$\Rightarrow$ dynamics of risk factor time series, a.o. predicting volatility, suitable for *market risk* and others.

Historical method/historical simulation



What is the idea?
If risk factors are i.i.d., then losses are are also i.i.d.?

$$F_n(l) = \frac{1}{n} \sum_{s=t-n+1}^{t} I_{\{\tilde{L}_s \leq l\}} = \frac{1}{n} \sum_{s=t-n+1}^{t} I_{\{l_{[t]}(\boldsymbol{X}_s) \leq l\}}$$
$$\rightarrow P(l_{[t]}(\boldsymbol{X}) \leq l) \equiv F_L(l)$$

Or: Look at the distribution of the losses over a historic time period, assume that in future losses will follow the same distribution.

How do we get VaR, single asset case?

1. Get the losses $L_t = -V \times r_t$, for principal V and return r_t
2. Order them, such that $L_{(1)} \leq \ldots L_{(T)}$
3. Get the empirical α quantile: $\mathrm{VaR}_L(\alpha) = L_{(k)}$, with $k = \lfloor \alpha n \rfloor$ (...)

How do we get ES:

1. Select those $T - k + 1$ losses, starting at VaR, $\mathrm{VaR}_L(\alpha) = L_{(k)} \leq \cdots \leq L_{(T)}$
2. Take average over these, $\mathrm{ES}_L(\alpha) = \frac{1}{T-k+1} \sum_{t=k}^{T} L_{(t)}$

Or ask Excel, or Python (with care!!!)... See Wikipedia Quantile
R1 vs R6

How precise can we estimate VaR?

$$\sigma^2(\mathrm{VaR}(\alpha)) = \frac{\alpha(1-\alpha)}{n f_L(\mathrm{VaR}(\alpha))^2}$$

Or, precision of VaR depends on

- Number of observations, losses, n (larger $\rightarrow$ more precision)
- Quantile (α more extreme $\rightarrow$ less precision)
- Density of the losses (if there are more losses close to VaR, easier to estimate)

Heij et al. (2004, §6.1.5)

What to do when you have a portfolio of assets?

1. Get the daily returns of underlying assets, $r_{i,t}$ with losses $L_{i,t} = -Vw_i r_{i,t}$ (using weights w_i, principal V)
2. Create the daily portfolio returns/losses $r^v_{p,t} = Vr^w_{p,t} = V\sum_i w_i r_{i,t}$, $L_{p,t} = -r^v_{p,t}$
3. Use portfolio losses to create VaR/ES, as in the single asset case

(possibly try in Excel, or Python)

Note: Careful, distinguish asset return $r_{i,t}$ from 'weighted asset return' $r^w_{i,t} = w_i r_{i,t}$ from 'portfolio asset return' $r^v_{i,t} = Vw_i r_{i,t}$: second return applies the portfolio weight $0 \leq w_i \leq 1$, last return applies also the principal V.

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Advantages of historical VaR:

- easy to compute and to understand
- no need to know correlations between assets in the portfolio

Disadvantages:

- Forecast of VaR never bigger than biggest recorded loss: "Driving a car while looking through the rear view mirror"
- (Less of a problem if stress period is included?)
- Historical data might be unavailable for all assets in portfolio
- Risk measures do not adapt to the current situation, i.e., this is an unconditional approach

If we assume that $\boldsymbol{r}_t \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma_r)$, multivariate normal, then

- the portfolio returns are also normal, with

$$r_{p,t}^w = \boldsymbol{W}'\boldsymbol{r}_t \sim \mathcal{N}(\boldsymbol{W}'\boldsymbol{\mu}, \boldsymbol{W}'\Sigma_r\boldsymbol{W}) \equiv \mathcal{N}(\mu_r^v, \sigma_r^{v2}),$$

  (see Lecture 1)
- one-day percentage-wise VaR is then simply

$$\mathrm{VaR}^w(1,\alpha) = -\mu_r^v + z_\alpha \sigma_r^v$$

  with z_α the α-quantile of the normal distribution, $-\mu_r^v$ the expected loss over one day.
- one-day monetary VaR is

$$\mathrm{VaR}^v(1,\alpha) = V\mathrm{VaR}^w(1,\alpha),$$

  multiplying by principal.

Suppose a portfolio, which is worth 1M Euros, has mean return of 6% p/a and volatility 17% p/a (obtained by e.g., variance-covariance formula)

Q: What is VaR(1, 97.5%)?

A:

- Mean loss over 1 day is approx. $-6\%/250 = -0.024\%$ (so almost zero)
- Volatility over 1 day is $17\%/\sqrt{250} = 1.08\%$ per day
- $\mathrm{VaR}^w(1d, 97.5\%) = -0.024\% + 1.96 \times 1.08\% = 2.08\%$
- In monetary units: $\mathrm{VaR}^v(1d, 97.5\%) = 1M \times 2.08\% = 20.8k$
- So in 2.5% of worst days, the loss of this portfolio will be 20.8k Euro or more

(year= 250 days; vol= $\sqrt{\text{var}}$; scale variance by number of days; scale volatility by $\sqrt{n}$)

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We have seen:

$$\begin{aligned}
\mathrm{ES}_\alpha &= E(L|L > \mathrm{VaR}_\alpha) \qquad \text{Expected Shortfall} \\
&= \int_{\mathrm{VaR}}^{\infty} z f_L(z|L > \mathrm{VaR}) dz \\
&= \frac{1}{P(L > \mathrm{VaR})} \int_{\mathrm{VaR}}^{\infty} z f_L(z) dz \\
&= \frac{1}{1-\alpha} \int_{\mathrm{VaR}}^{\infty} z f_L(z) dz
\end{aligned}$$

What is this?

NB: $\alpha = 0.975$, for ES(97.5%)...

Expected Shortfall: Standard normal case II

Suppose $L \sim \mathcal{N}(0,1)$. Then

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right)$$
$$\phi'(z) = -\frac{z}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}z^2\right) = -z\phi(z)$$

Hence

$$\mathrm{ES}_\alpha \overset{N(0,1)}{=} \frac{1}{1-\alpha}\left[-\phi(z)\right]_{\mathrm{VaR}}^{\infty} = \frac{1}{1-\alpha}\left(0 + \phi(\mathrm{VaR})\right)$$
$$= \frac{1}{1-\alpha}\phi(\Phi^{\leftarrow}(\alpha))$$

where $\mathrm{VaR} \equiv \Phi^{\leftarrow}(\alpha) = \Phi^{-1}(\alpha)$, the z_α quantile value.

Suppose $L \sim \mathcal{N}(\mu, \sigma^2)$ instead.
Then use a location-scale change-of-variables technique:

$$\mathrm{ES}_{\mu,\sigma^2,\alpha} \overset{N(\mu,\sigma^2)}{=} \mu + \sigma \mathrm{ES}_\alpha$$

(Works for any location-**scale** family of distributions)
MFE §2.3.2

In the Student-t case, for the standard distribution with scale $c = 1$, the equation is less nice:

$$\mathrm{ES}_{\nu,\alpha} \overset{t(\nu,0,1)}{=} \frac{1}{1-\alpha} f_\nu(F_\nu^{\leftarrow}(\alpha)) \left(\frac{\nu + (F_\nu^{\leftarrow}(\alpha))^2}{\nu - 1} \right)$$

To get to a Student-t distribution with standard deviation s and expected return μ,

$$\mathrm{ES}_{\nu,\mu,s^2,\alpha} = \mu + s\sqrt{\frac{\nu-2}{\nu}}\mathrm{ES}_{\nu,\alpha}$$

MFE (2.25)

Warning: Watch out for the factor adapting for the Student-t variance, book is unclear (see below)
Hint: Check your results with a simulation...

What could I lose over h days?

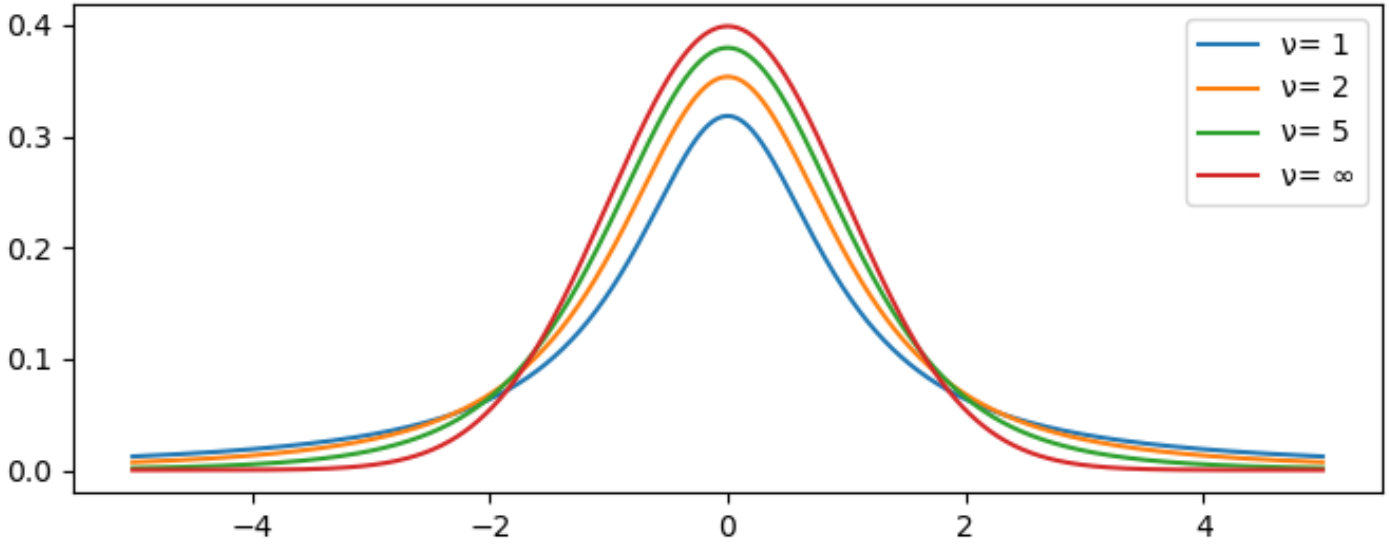
1. Simple answer: $\mathrm{VaR}(h, \alpha) = \sqrt{h}\mathrm{VaR}(1, \alpha)$ (correct if returns are uncorrelated, mean roughly zero)
2. More involved answer: $\mathrm{VaR}(h, \alpha) = h\mu_1 + z_\alpha \sqrt{h}\sigma_1$, if we assume the normal VaR, and no correlation between days
3. Even more involved answer: $\mathrm{VaR}(h, \alpha) = \mu_h + z_\alpha \sigma_h$, if we assume the normal VaR and look at h-day mean loss and h-day variance.

For ES: Use h-day mean loss/variance, normal ES formula.

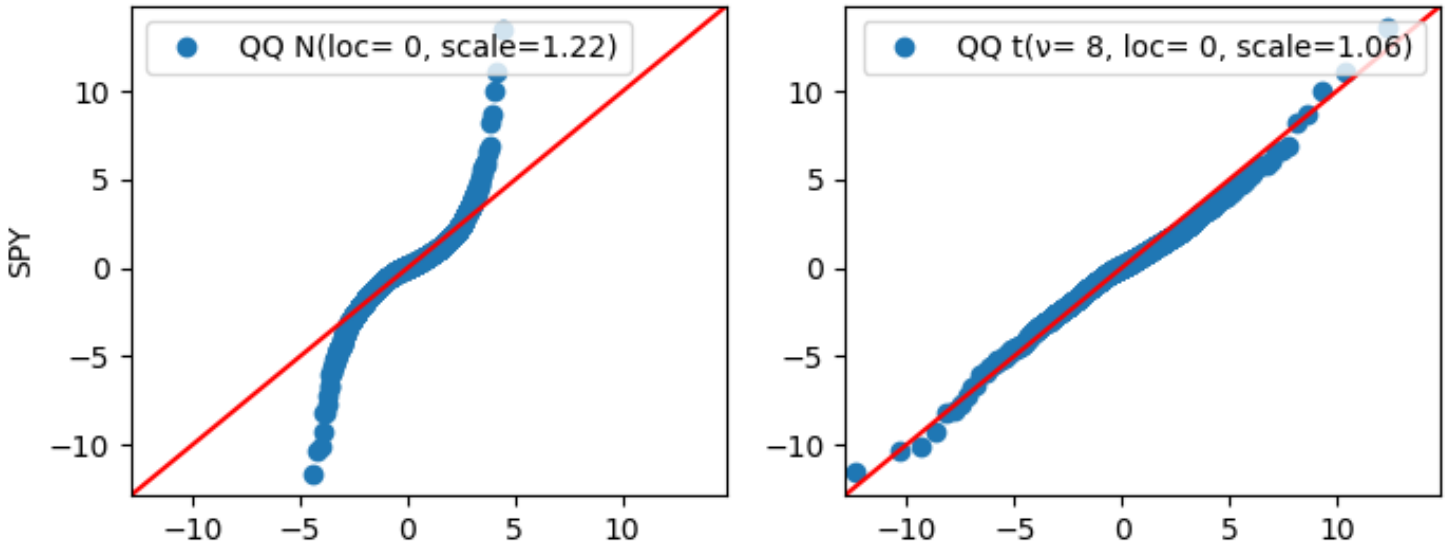
Q: Calculate VaR($h = 10, \alpha = 97.5\%$)
A:

- 10-day mean loss$= 10 \times \mu_1 = -10 \times \frac{6\%}{250} = -0.24\%$
- 10-day volatility$= \sqrt{10} \times 17\%/\sqrt{250} = 3.4\%$
- $\mathrm{VaR}(10, 97.5\%) = -0.24 + 1.96 \times 3.4 = 6.42\%$, or 64200 Euro
- Simple answer, assuming $\mu_1 = 0$:
  $\mathrm{VaR}(10, 97.5\%) = \sqrt{10} \times 2.08\% = 6.58\% \equiv 65800$ Euro

The eternal question: Normal or Student-t distribution?



- Fin. returns are typically heavy tailed
- Hence replace Normal-VaR by Student-t VaR?
- Using low df$=\nu$
- with QQ-plot to decide on ν
- But careful with effect of variance, volatility clustering



SPY 2000-2026/2 QQ-plot

Careful with scale:

- With normal, scale $c \equiv \sigma$
- With Student-t, variance $\sigma^2 = c^2 \frac{\nu}{\nu-2}$, so scale $c \equiv \sigma / \sqrt{\frac{\nu}{\nu-2}}$

WARNING: In the VaR and ES formulas, you need the scale parameter, not the volatility!

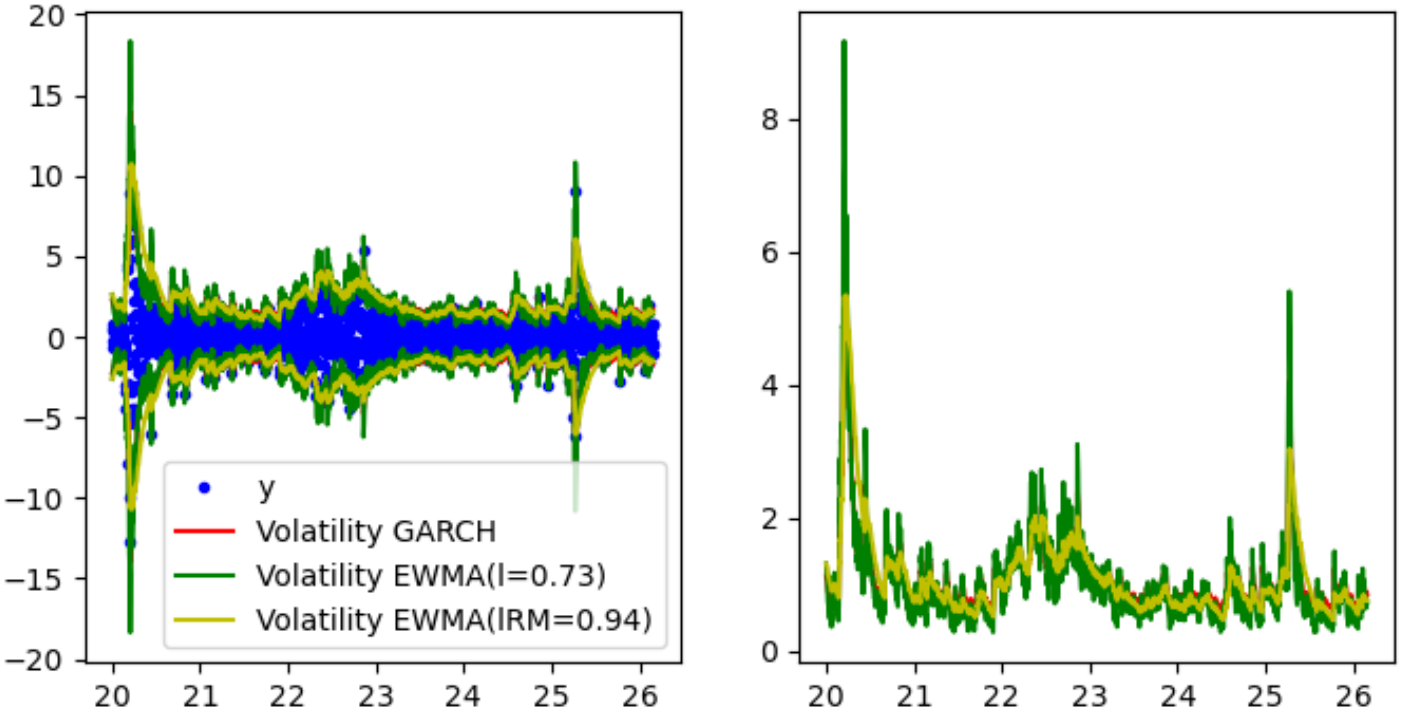
Now repeat, using conditional loss $L_{t+1}|\mathcal{F}_t$ instead of unconditional loss L_{t+1}

- So we need to obtain conditional VaR and ES (which now depend on time t):

$$\mathrm{VaR}_{t,\alpha} = F^{\leftarrow}_{L_{t+1}|\mathcal{F}_t}(\alpha)$$
$$\mathrm{ES}_{t,\alpha} = E(L_{t+1}|L_{t+1} > \mathrm{VaR}_{t,\alpha}, \mathcal{F}_t)$$

- In Var-Cov method: Change to σ_t, time varying volatility.
- Historical simulation method: Use **Filtered Historical Simulation**, see (McNeil, Frey, and Embrechts, 2015, §9.2.4)/below

Q: In Var-Cov, why do we not change from μ to μ_t?



S&P 500 2020-2026/2 Volatility plot

How important is the non-constant volatility?

Simple possibility: Exponentially Weighted Moving Average of variance

If we assume $u_t \sim \cdot(0, \sigma_t^2)$ (e.g. $\mathcal{N}$ or t, or non-specified)

$$\sigma_t^2 = \sum_{i=1}^{m} \alpha_i u_{t-i}^2$$

$$\sum_{i=1}^{m} \alpha_i = 1$$

Advantage:

- Simple
- Current observations may be more heavily weighted
- Weights of past observation may 'decrease' over time

E.g. $\alpha_i = \frac{1}{m} I_{1 \leq i \leq 60}$, weigh only last $m = 60$ days

How to choose α alternatively?
EWMA filter:

$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)u_{t-1}^2 \Leftrightarrow \alpha_i \qquad = \lambda^{i-1}(1-\lambda)$$

- Current observation gets weight $(1-\lambda)$
- Weights dropping off exponentially
- Choice of $\lambda = 0.94$: RiskMetrics, for daily volatility forecasting
- Sometimes $\lambda = 0.97$, reduces cyclicality of risk measures, less reactive behaviour, more stable
- Or, for truly advanced (but not necessarily better...) usage: Optimise

MFE §4.2.5, Equation (4.47)

The EWMA can be seen as a special case of the GARCH(1,1) filter (Bollerslev, 1986),

$$u_t|\mathcal{F}_{t-1} = \mu_t + \sigma_t z_t \qquad a_t \equiv u_t - \mu_t \qquad z_t \sim \mathcal{N}(0,1)$$
$$\sigma_t^2 = \alpha_0 + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

with $\alpha_0 > 0, 0 \leq \alpha_1 \leq 1, 0 \leq \alpha_1 + \beta_1 \leq 1$
MFE §4.2

Expected variance

For GARCH(1,1), the unconditional expectation/long term variance is

$$E\sigma_t^2 = V_L = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}$$

Assume we estimated

$$
\begin{aligned}
u_t|\mathcal{F}_{t-1} &= \mu_t + \sigma_t z_t \\
\sigma_t^2 &= \alpha_0 + \alpha_1(u_{t-1} - \mu_{t-1})^2 + \beta_1 \sigma_{t-1}^2 \\
\mu_t &= 0, \alpha_0 = 0.000002, \alpha_1 = 0.13, \beta = 0.86
\end{aligned}
$$

Q:

- What is the long run variance?
- What is the long run volatility per day?
- If current volatility is 1.6%, last return is 1%, what would be tomorrow's variance and volatility?
- What is current VaR at the 97.5% level, for the investment of 1M EUR? What would it be tomorrow?

A:

- $E\sigma_t^2 = V_L = \frac{\alpha_0}{1-\alpha_1-\beta_1} = \frac{0.000002}{1-0.13-0.86} = .0002$
- Long run volatility$= \sqrt{V_L} = 0.01414 \equiv 1.4\%$
- $\sigma_{t+1}^2 = \alpha_0 + \alpha_1(u_t - \mu_t)^2 + \beta_1\sigma_t^2 =$ $0.000002 + 0.13 \times (.01 - 0)^2 + 0.86 \times .016^2 = .00023516$, so tomorrow's volatility is $s_{t+1} = \sqrt{\sigma_{t+1}^2} = 1.53\%$
- $\mathrm{VaR}(1d, 0.975) = 0.016 \times 1.96 \times 1M = 31360$ EUR; tomorrow we would get $\mathrm{VaR}(1d, 0.975) = 0.0153 \times 1.96 \times 1M = 30055$ EUR (or 29988?)

Q: Tomorrow's VaR, is it 30055 EUR or 29988 EUR? Which one is correct at the exam?

The Generalized AutoRegressive Conditional Heteroskedasticity(1,1) model can be generalized to allow for more lags:

$$u_t|\mathcal{F}_{t-1} = \mu_t + \sigma_t z_t \qquad a_t \equiv u_t - \mu_t \qquad z_t \sim \mathcal{N}(0,1)$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^{p} \alpha_i a_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2$$

with $\alpha_0 > 0, \alpha_i \geq 0, \beta_j \geq 0$. If $\sum_i \alpha_i + \sum_j \beta_j < 1$, the GARCH($p, q$) is covariance stationary.

Notes:

- Higher orders are harder to estimate
- They tend to add little practical value compared to GARCH(1,1)

MFE §4.2.2

See McNeil, Frey, and Embrechts (2015, §9.2.4).
What is it? Three options:

1. Use ARMA-GARCH(-t), use model predictions to get $\hat{\mu}_{t+1}, \hat{\sigma}_{t+1}$, and standard method for VaR/ES
2. Use ARMA-GARCH(-t), but extract historical residuals z_t from model. Find empirical quantile $q_z(\alpha_{\mathrm{VaR}})$, and use this instead of $\Phi^{-1}(\alpha_{\mathrm{VaR}})$, apply $\hat{\mu}_{t+1}, \hat{\sigma}_{t+1}$. Similar for ES
3. Use EWMA instead of model to get variance, use historic residuals as in 2.

For multi-asset portfolio's, prefer dynamic corr/cov!
Try EWMA-like covariance between a and b:

$$\sigma_{a,t}^2 = \lambda_a \sigma_{a,t-1}^2 + (1-\lambda_a) r_{a,t-1}^2$$
$$\sigma_{b,t}^2 = \lambda_b \sigma_{b,t-1}^2 + (1-\lambda_b) r_{b,t-1}^2$$
$$\sigma_{ab,t} = \lambda_{ab} \sigma_{ab,t-1} + (1-\lambda_{ab}) r_{a,t-1} r_{b,t-1}$$

$\Rightarrow$ no guarantee to get positive definite covariance matrix

$$\Sigma_t = \begin{pmatrix} \sigma_{a,t}^2 & \sigma_{ab,t} \\ \sigma_{ab,t} & \sigma_{b,t}^2 \end{pmatrix}$$

Or: We want

$$X_t | \mathcal{F}_{t-1} \sim \cdot(0, \Sigma_t)$$
$$\Sigma_t = \Delta_t P_t \Delta_t$$

with time varying volatility Δ_t, and time varying correlation P_t, positive definite always.

Simple solution: Constant Conditional Correlation model

1. Estimate GARCH on $r_{i,t}$, all assets separately, to get $\sigma^2_{i,t}$
2. Construct volatility $\Delta_t = \text{diag}(\sigma_{i,t}, i = 1, \ldots, k)$
3. Get *devolatilised* process $Y_t = \Delta_t^{-1} X_t$
4. Calculate correlation $P \equiv \mathrm{Corr}(Y_t)$
5. Construct $\Sigma_t = \Delta_t P \Delta_t$

MFE §14.2

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MFE §14.2

Advantage:

- Nice and simple to estimate
- Combines well with VaR-Cov method for VaR and ES estimation

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MFE §14.2

Disadvantage:

- Correlation indeed is fixed, also in times of stress
- So covariances increase during high volatility in the markets!

More advanced solution: Dynamic Conditional Correlation model

1. Estimate GARCH on $r_{i,t}$, all assets separately, to get $\sigma^2_{i,t}$
2. Construct volatility $\Delta_t = \text{diag}(\sigma_{i,t}, i = 1, \ldots, k)$
3. Get *devolatilised* process $Y_t = \Delta_t^{-1} X_t$
4. Calculate long-run correlation $P_0 \equiv \mathrm{Corr}(Y_t)$
5. Filter $Q_t = (1 - \alpha^\rho - \beta^\rho) P_0 + \alpha^\rho Y_{t-1} Y_{t-1} + \beta^\rho P_{t-1}$, $P_t = \Delta(Q_t)^{-1} Q_t \Delta(Q_t)^{-1}$, where $\Delta(Q_t)$ is the diagonal matrix with the volatility of Q_t on the diagonal.
6. Optimise the previous step over α^ρ, β^ρ
7. Construct $\Sigma_t = \Delta_t P_t \Delta_t$

Warning: Not too simple... Maybe use a package for it!

Step back a moment: What are we doing?

1. We have a portfolio, $V_t = \boldsymbol{w}_t' \boldsymbol{p}_t$, or $r_{pt} = \boldsymbol{w}_t' \boldsymbol{r}_t$
2. We're looking for VaR and/or ES of portfolio
3. Options:
    1. Use portfolio returns r_{pt}, construct VaR/ES using univariate methods
    2. Use asset returns $\boldsymbol{r}_t$, construct VaR/ES using multivariate methods from $\mu_t = \boldsymbol{w}_t \boldsymbol{\mu}_t, \sigma_{pt}^2 = \boldsymbol{w}_t' \Sigma_t \boldsymbol{w}_t$

Last option: Plug in CCC/DCC...

Study the following chapters/sections of the book:

- Chapter 4, section 4.2
- Chapter 9:
  - 9.2 (except EVT-based estimators, pp 348-349)
  - 9.3: subsections 9.3.1, 9.3.2, 9.3.4
- Section 9.3, and especially subsection 9.3.4 will serve you as a guidance for your first assignment: comparison of various RM models
- Section 14.2, for CCC/DCC, skipping e.g. BEKK

Bollerslev, Tim (1986). "Generalized Autoregressive Conditional Heteroskedasticity". In: *Journal of Econometrics* 31.3, pp. 307–327. DOI: 10.1016/0304-4076(86)90063-1.

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McNeil, Alexander J., Rüdiger Frey, and Paul Embrechts (2015). *Quantitative Risk Management: Concepts, Techniques, and Tools*. Revised. Princeton: Princeton University Press. ISBN: 978-0-691-16627-8.